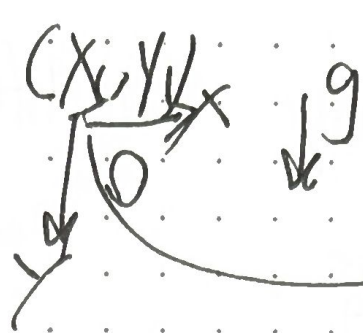


Brachistochrone Problem



quickest path?

$$I = \int_0^t dt = \int_A^B \frac{ds}{v}$$

$\leftarrow$ velocity

$$\frac{1}{2}mv^2 = mgy \quad \text{(conservation of energy)}$$

$$v = \sqrt{2gy}$$

$$ds^2 = dx^2 + dy^2 \quad \frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$I = \int_{x_1}^{x_2} \frac{\sqrt{1 + (y')^2}}{\sqrt{2gy}} dx$$

$$= \frac{1}{\sqrt{2g}} \int_{x_1}^{x_2} \sqrt{\frac{1 + (y')^2}{y}} dx$$

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= \sqrt{1 + (y')^2} dx$$

$$\boxed{y' = \frac{dy}{dx} \text{ or } \frac{\partial y}{\partial x}}$$

$$F = \sqrt{\frac{1 + (y')^2}{y}} = F(y, y') \rightarrow \text{Beltrami identity}$$

$$F - y' \frac{\partial F}{\partial y'} = \text{const}$$

$$\sqrt{\frac{1 + (y')^2}{y}} - y' \cdot \frac{1}{\sqrt{y}} \cdot \frac{1}{2} (1 + (y')^2)^{-\frac{1}{2}} \cdot 2y' = C$$

$$\sqrt{\frac{1 + (y')^2}{y}} - \frac{(y')^2}{\sqrt{y(1 + (y')^2)}} = C$$

$$\rightarrow 1 + (y')^2 - (y')^2 = C^2 \sqrt{y(1 + (y')^2)^2}$$

$$1 = C^2 \sqrt{y(1 + (y')^2)}$$

$$\frac{1}{C^2} = C_1 = y + y(y')^2$$

$$\frac{dy}{dx} = \sqrt{\frac{C_1 - y}{y}}$$

separation of variable

$$dx = \sqrt{\frac{y}{C_1 - y}} dy$$

$$\int dx = \int \sqrt{\frac{y}{C_1 - y}} dy$$

$$X + C_2 = \int \frac{y}{C_1 - y} dy \quad \text{substitute } y = C_1 \sin^2 \theta \quad 0 < \theta < \frac{\pi}{2}$$

$$dy = 2C_1 \sin \theta \cos \theta d\theta$$

$$= \int \frac{C_1 \sin^2 \theta}{C_1 - C_1 \sin^2 \theta} \cdot 2C_1 \sin \theta \cos \theta d\theta$$

$$= \int \frac{\sin \theta}{\cos \theta} \cdot 2C_1 \sin \theta \cos \theta d\theta = 2C_1 \int \sin^2 \theta d\theta$$

$$= 2C_1 \int \frac{1}{2} (1 - \cos 2\theta) d\theta$$

$$= C_1 \left(\theta - \frac{1}{2} \sin 2\theta \right)$$

$$C_1 \theta - \frac{C_1}{2} \sin 2\theta - C_2$$

$$y = C_1 \sin^2 \theta = \frac{C_1}{2} (1 - \cos 2\theta)$$

C_1, C_2 can be found via boundary conditions

$$X = C_1 \theta - \frac{C_1}{2} \sin 2\theta - C_2$$

$$Y = C_1 \sin^2 \theta = \frac{C_1}{2} (1 - \cos 2\theta)$$

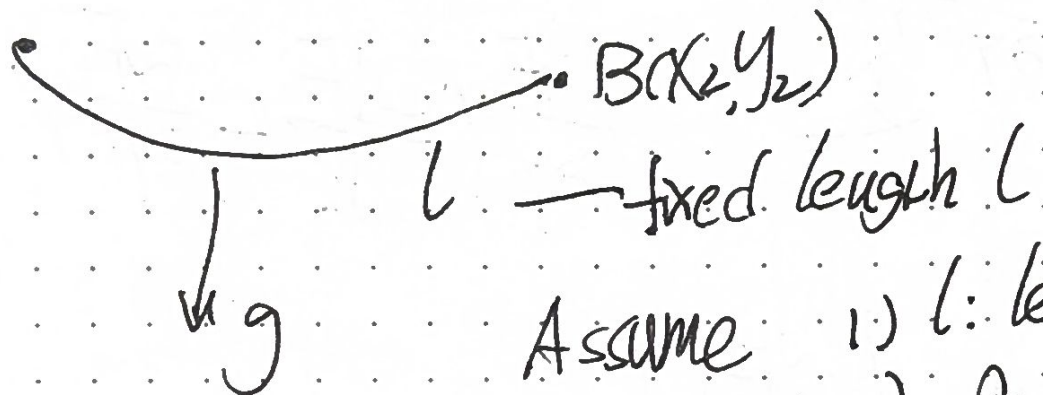
} cycloid

The hanging chain problem (catenary problem)

CATENARY PROBLEM: what is the shape of a hanging chain/rope pulled by gravity?

$$A = (x_1, y_1)$$

$$B = (x_2, y_2)$$



fixed length l

Assume 1) l : length is constant
2) ρ, A : constant

Minimize potential energy

$$I = \int_A^B mgy ds$$

constraint: l

$$\frac{ds}{dx} = \sqrt{1 + (y')^2}$$

$$J = \int_A^B ds = l$$

that's

$$I = \int_{x_1}^{x_2} mgy \sqrt{1 + (y')^2} dx$$

$$J = \int_{x_1}^{x_2} \sqrt{1 + (y')^2} dx = l$$

we wish to minimize I subject to constraint l

we need Lagrange multiplier

$$\delta(I + \lambda J) = 0 \quad \lambda: \text{Lagrange multiplier a unknown constant}$$

$$\delta \left(\int_{x_1}^{x_2} (mgy + \lambda) \sqrt{1 + (y')^2} dx \right) = 0$$

$$\delta \int_{x_1}^{x_2} (mgy + \lambda) \sqrt{1 + (y')^2} dx = 0 \quad F = F(y, y')$$

apply Euler-Lagrange equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\frac{dF}{dx} = \frac{\partial F}{\partial y} \frac{dy}{dx} + \frac{\partial F}{\partial y'} \frac{dy'}{dx}$$

$$\frac{\partial F}{\partial y} y' - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) y' = 0$$

$$\frac{\partial F}{\partial y} y' = \frac{dF}{dx} - \frac{\partial F}{\partial y'} y''$$

$$\frac{dF}{dx} - \frac{\partial F}{\partial y'} y'' - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) y' = 0$$

$$\frac{\partial F}{\partial y'} y'' + \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) y' - \frac{dF}{dx} = 0$$

it is just $\frac{d}{dx} \left(y' \frac{\partial F}{\partial y'} - F \right) = 0$

So $y' \frac{\partial F}{\partial y'} - F = \text{constant}$.

↳ Beltrami identity

why F is not dependent on x

real: $F = (mgy + \lambda) \sqrt{1 + (y')^2}$

$$y' (mgy + \lambda) \cdot \frac{1}{2} \cdot (1 + (y')^2)^{-\frac{1}{2}} \cdot 2y' - (mgy + \lambda) (1 + (y')^2)^{-\frac{1}{2}} = \text{const}$$

$$\frac{(mgy + \lambda)}{\sqrt{1 + (y')^2}} (y')^2 - (mgy + \lambda) \sqrt{1 + (y')^2} = \text{const}$$

$$\text{thats } -\frac{mgy + \lambda}{\sqrt{H(y)^2}} = \text{const.} \Rightarrow \frac{mgy + \lambda}{\sqrt{1 + (y')^2}} = C$$

$$y' = \frac{dy}{dx} = \sqrt{\left(\frac{mgy + \lambda}{C}\right)^2 - 1} \quad \text{separate variables}$$

$$dx = \frac{dy}{\left(\frac{mgy + \lambda}{C}\right)^2 - 1}$$

$$\int dx = \int \frac{dy}{\left(\frac{mgy + \lambda}{C}\right)^2 - 1}$$

$$\text{since } \frac{mgy + \lambda}{C} = \cosh \eta$$

$$x + C_2 = \frac{C}{mg} \int \frac{\sinh \eta}{(\cosh^2 \eta - 1)} d\eta \rightarrow \sinh \eta$$

$$\frac{mg dx}{C} = \cosh \eta d\eta$$

$$x + C_2 = \frac{C}{mg} \eta$$

$$dy = \frac{C \sinh \eta d\eta}{mg}$$

$$\frac{mgy + \lambda}{C} = \cosh \eta \quad \eta = \cosh^{-1} \left(\frac{mgy + \lambda}{C} \right)$$

$$x + C_2 = \frac{C}{mg} \cosh^{-1} \left(\frac{mgy + \lambda}{C} \right)$$

$$\frac{mgy + \lambda}{C} = \cosh \left(\frac{mg}{C} (x + C_2) \right)$$

$$y = \frac{C}{mg} \cosh \left(\frac{mg}{C} (x + C_2) \right) - \frac{\lambda}{C}$$

real constraint: $\int_{x_1}^{x_2} \sqrt{1 + (y')^2} dx = l$

$$y = \frac{c_1}{mg} \cosh\left(\frac{mg}{c_1}(x + c_2)\right) - \frac{1}{c_1}$$

$$\begin{aligned} y' = \frac{dy}{dx} &= \frac{c_1}{mg} \sinh\left(\frac{mg}{c_1}(x + c_2)\right) \cdot \frac{mg}{c_1} \\ &= \sinh\left(\frac{mg}{c_1}(x + c_2)\right) \end{aligned}$$

set $\int_{x_1}^{x_2} \sqrt{1 + \sinh^2\left(\frac{mg}{c_1}(x + c_2)\right)} dx = l$ $1 + \sinh^2 x = \cosh^2 x$

$$\int_{x_1}^{x_2} \cosh\left(\frac{mg}{c_1}(x + c_2)\right) dx = l$$

$$\frac{c_1}{mg} \sinh\left(\frac{mg}{c_1}(x + c_2)\right) \Big|_{x_1}^{x_2} = l$$

$$\frac{c_1}{mg} \left[\sinh\left(\frac{mg}{c_1}(x_2 + c_2)\right) - \sinh\left(\frac{mg}{c_1}(x_1 + c_2)\right) \right] = l$$

need numerical solution

$$\begin{aligned} x &= x_1, y = y_1 \\ x &= x_2, y = y_2 \end{aligned}$$